

Laura Bussi - Curriculum Vitae

2nd October 2026

Status: Lecturer,
University of Amsterdam

Fields: Logic, Model Checking, Artificial Intelligence

Languages: Italian: Native
English: C1 (working language)
German: B1 (basic communication)
Dutch: A1 (beginner)

Scholar: [Google Scholar profile](#)

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Profile

Researcher in formal methods and artificial intelligence, focusing on spatial and spatio-temporal reasoning, model checking, and verification of learning-based systems. My work develops logical and algorithmic foundations for trustworthy AI, with an emphasis on **tool-supported** methods and safety-critical applications in domains such as medical imaging.

Research interests

Formal methods for reasoning about large structured systems
Spatial and topological reasoning
Neuro-symbolic artificial intelligence
Trustworthy and explainable machine learning

Work experience

09/2025-now	Lecturer <i>University of Amsterdam, Informatics Institute</i> Teaching in the Master Programme in Software Engineering, with continued research activity in systems verification and supervision of Master's students.	📍 Amsterdam, Netherlands
03/2025-08/2025	Postdoctoral fellow <i>University of Luxembourg, SnT Center</i> Verification of neural networks using temporal logics and formal methods.	📍 Luxembourg, Luxembourg
02/2024-02/2025	Postdoctoral fellow <i>National Research Council, Formal Methods and Tools Lab</i> Development of new tools and techniques based on Artificial Intelligence and Spatial Logics for medical imaging (fellowship). Project: PRIN Tuscany Health Ecosystem (CUP I53C22000780001)	📍 Pisa, Italy

Education

11/2020-05/2024	PhD in Computer Science <i>University of Pisa</i> Advisors: Prof. Fabio Gadducci, Dott. Vincenzo Ciancia PhD thesis: Model checking properties with identity binding in space-time	📍 <i>Pisa, Italy</i>
11/2018-10/2020	Master's Degree in Computer Science <i>University of Pisa</i> Software: programming, principles and technologies Master's Thesis: A GPU implementation of the spatial model checker VoxLogicA	📍 <i>Pisa, Italy</i>
09/2015-11/2018	Bachelor's Degree in Computer Science <i>University of Pisa</i> Bachelor's Thesis: Polyadic soft constraints	📍 <i>Pisa, Italy</i>

Publications

Journal article 2026	Spatio-temporal Model Checking with VoxLogicA Bussi L., Ciancia V., Gadducci F., Massink M. <i>Journal of Logic, Language, and Information</i> , Springer Nature, Volume 35, 83–108
Journal article 2026	Weak Simplicial Bisimilarity and Minimisation for Polyhedral Model Checking Bezanishvili N., Bussi L., Ciancia V., Gabelaia D., Jibladze M., Latella D., Massink M. <i>Logical Methods in Computer Science</i> , Volume 22, Issue 1
Preprint 2025	ABS: Enforcing Constraint Satisfaction On Generated Sequences Via Automata-Guided Beam Search Bussi L., Collura V., Cordy M., Giunchiglia E., Tit K. arXiv:2506.09701
Conference paper 2024	Towards Hybrid-AI in Imaging using VoxLogicA Belmonte G., Bussi L., Ciancia V., Latella D., Massink M. <i>Leveraging Applications of Formal Methods, Verification and Validation. Software Engineering Methodologies</i> , Volume 15222, 205–221
Conference paper 2024	Logics of Polyhedral Reachability Bezhanishvili N., Bussi L., Ciancia V., Duque D.F., Gabelaia D. <i>Advances in Modal Logic - AiML2024</i> , to appear
Journal article 2023	A Toolchain for Strategy Synthesis with Spatial Properties Basile D., Bussi L., Ciancia V., Ter Beek M. <i>International Journal on Software Tools for Technology Transfer</i> , Springer Nature, Volume 25, 641–658
Conference paper 2023	A Spatial Logic with Time and Quantifiers Bussi L., Ciancia V., Gadducci F. <i>DaLi: Dynamic Logic - New Trends and Applications</i> , Springer, 1–19
Conference paper 2022	On Binding in the Spatial Logic for Closure Spaces Bussi L., Ciancia V., Gadducci F., Latella D., Massink M. <i>International Symposium on Leveraging Applications of Formal Methods</i> , Springer International Publishing, 479–497
Conference paper 2022	Soft Concurrent Constraint Programming with Local Variables Bussi L., Gadducci F., Santini F. <i>International Conference on Coordination Languages and Models</i> , Springer Nature Switzerland, 159–177
Conference paper 2021	Querying medical imaging datasets using spatial logics (position paper) Belmonte G., Broccia G., Bussi L., Ciancia V., Latella D., Massink M. <i>Advances in Model and Data Engineering in the Digitalization Era: MEDI 2021</i> , Springer International Publishing, 285–301
Conference paper 2021	Towards Model Checking Video Streams Using VoxLogicA on GPUs Bussi L., Ciancia V., Gadducci F., Latella D., Massink M. <i>International Symposium: From Data to Models and Back</i> , Springer International Publishing, 78–90

Conference paper
2021

Towards a Spatial Model Checker on GPU Bussi L., Ciancia V., Gadducci F. International Conference on Formal Techniques for Distributed Objects, Components, and Systems, Springer International Publishing, 188–196

Conference paper
2019

Polyadic Soft Constraints Bussi L., Gadducci F., Santini F. The Art of Modelling Computational Systems: A Journey from Logic and Concurrency to Security and Privacy, Springer International Publishing, 241–257

Grants and awards

2026

Paolo Gentilini award for applications of logic to biomedical science  *Rome, Italy*
Italian Association for Logic and its Applications

2024

FoSSaCS 2024 Distinguished reviewer award  *Luxembourg, Luxembourg*
University of Luxembourg

2023

CNR Grant for Young Mobility  *Pisa, Italy*
National Research Council
The awarded project is a research visit in Georgia to work on axiomatisation of modal logic with reachability on polyhedra.

Research periods abroad

07/2025

Department of Electrical and Electronic Engineering  *London, United Kingdom*
Imperial College London
Presentation and study on constrained output generation using neural networks with Dr. Eleonora Giunchiglia.

09-10/2023

Andrea Razmadze Mathematical Institute  *Tbilisi, Georgia*
Tbilisi State University
Study on polyhedral completeness of modal logic with reachability with Prof. David Gabelaia.

02-03/2023

Institute for Logic, Language and Computation  *Amsterdam, Netherlands*
University of Amsterdam
Study on polyhedral completeness of modal logic with reachability with Prof. Nick Bezhanishvili.

Invited talks

26/06/2025

International Workshop on Topological Methods in Logic VII  *Tbilisi, Georgia*
Tbilisi State University
Polyhedral Logic with Reachability: from semantics to minimization.

Teaching experience

2025-2026	Lecturer <i>University of Amsterdam</i> Software Specification, Verification and Testing Model-based Design of Cyber-Physical Systems Software Evolution DevOps and Cloud Based Software Master Programme in Software Engineering	📍 Amsterdam, Netherlands
2020-2023	Teaching Assistant <i>University of Pisa</i> Advanced Programming, Master Programme in Computer Science and Networking	📍 Pisa, Italy
10/2019-12/2019	Teaching Assistant <i>University of Pisa</i> Logic for Programming, Bachelor Programme in Computer Science	📍 Pisa, Italy
06/2019-07/2019	Teacher <i>University of Pisa</i> Introduction to Micro:Bit programming, Summer course for high school students	📍 Pisa, Italy
11/2018-04/2019	Counselor <i>University of Pisa</i> Logic for programming, Bachelor Programme in Computer Science	📍 Pisa, Italy

Conference organization

AEC Chair	Symposium on Model Checking of Software, 2026
AEC Chair	European joint conferences on Theory And Practice of Software, 2025
PC Member	International Symposium On Formal Methods, Artifact Evaluation Track, 2024
PC Member	International Conference on Foundations of Software Science and Computation Structures, Artifact Evaluation Track, 2024
PC Member	International Conference on Formal Methods in Software Engineering, Artifact Evaluation Track, 2024
PC Member	International Conference on Formal Methods in Software Engineering, Artifact Evaluation Track, 2023
Local organiser	ItaCa: Italian Category Theorists workshop, Pisa, 2022

Tools and programming languages

Projects	Main developer of VoxLogicA-GPU , a high-performance voxel-based image analyser for medical imaging, leveraging GPU acceleration for practical use in healthcare contexts. Contributor to the VoxLogicA project (Kripke structures extension and VoxLogicA2.0).
Languages	C, C++, .NET languages, Java, Python, OCaml, Haskell, OpenCL, Prolog (my GitHub repositories here).

Other activities

Reviewer	Reviewer for various journals and conferences, such as Fuzzy Sets and Systems, the Journal of Systems and Software, NeurIPS, NeSy, and others.
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